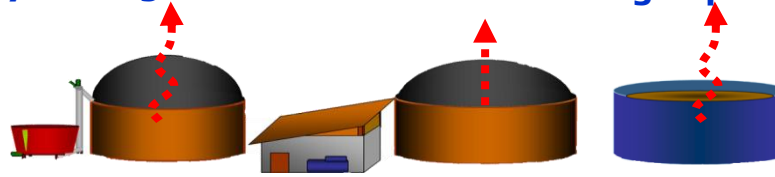


**Methane
yield &
quality of
substrate**

Biomass yield and ensiling

**Process optimisation
quality of biogas residues & Trace gas emissions
from biogas plant**



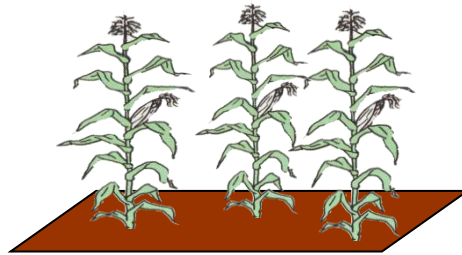
**Methane production and residue storage
Biogas plant**

**Fertilization
Biogas residues**

NH₃ emission

**Greenhouse
gas
emission
Soil fertility**

Biogas Production System



NO₃⁻

**Biomass accumulation &
N-efficiency of energy crops**

Biomass accumulation

**N load & N concentration
in leachate**

N₂O

NH₃

CH₄

NO₃

**Systems
evaluation**

**Systems
modeling**

**Economic
evaluation**

**Life cycle
assessment**